

Wrocław University of Science and Technology

Faculty of Electronics, Photonics and Microsystems

Field of study: Electronics

Speciality: Advanced Applied Electronics

Master Thesis

Title of Thesis:

**Design and analysis of free-space optical
data link using SFP modules**

AUTHOR:

Kacper Ugorny

SUPERVISOR:

dr inż. Aleksander Głuszek

WROCŁAW 2026

Streszczenie

W niniejszej pracy przedstawiono realizację autorskiego, szybko przepustowego łącza optycznego w wolnej przestrzeni z aktywną kompensacją nachylenia wiązki. Na jego potrzebę opisano architekturę i wykonanie autorskich urządzeń nadawczo-odbiorczych, wykonanych z przystępnych cenowo i szeroko dostępnych elementów na rynku, między innymi modułów SFP. Urządzenia te zaprojektowano z myślą o szerokim wykorzystaniu technologii druku 3D. Omówione zostały również badania tego łącza optycznego, porównano wydajność przy wyłączonej i włączonej kompensacji wiązki oraz wpływ różnych warunków atmosferycznych na komunikację optyczną w wolnej przestrzeni. Na koniec przedstawiono rozważania na temat przyszłego rozwoju projektu.

Abstract

This paper presents the implementation of a custom, high-throughput free-space optical link featuring active beam tilt compensation. To support this system, the architecture and construction of custom transceivers are described. These devices were built using affordable, commercially off-the-shelf components, including SFP modules, and were designed with a strong focus on utilizing 3D printing technology. Furthermore, the testing and evaluation of the optical link are discussed, performance of the link with and without beam compensation was compared and influence of weather conditions on free space optical communication. At the end considerations for the future development of the project are showcased.

Contents

1. List of Acronyms	6
2. Introduction	7
2.1 Background	7
2.2 Motivation	8
2.3 Thesis Objectives	8
2.4 Scope and Structure	8
3. Theoretical Background and State of the Art	10
3.1 Current Research Directions	10
3.2 Principles of FSO Communication	10
3.3 Impact of envirometnal factors on light propagation	13
3.4 SFP Modules and Digital Diagnostic Monitoring (DDM)	13
3.5 Existing Beam Steering Techniques	14
4. System Architecture and Hardware Design	16
4.1 Transceiver and Link Overview	16
4.2 Mechanical and Optical Design	17
4.3 Controller Design	24
5. Active Control Algorithm	27
5.1 DDM Data Extraction	27
5.2 Connection Seeking Algorithm (Spiral Search)	28
5.3 Power aligment algorithm	29
5.4 Control Algorithm Design	30
5.5 Misalignment Compensation	32
6. Experimental Setup and Measurements	33
6.1 Experimental Setup	33
6.2 Test Scenarios and Measurement metrics	35
6.3 Link throughput measurement	35
6.4 Baseline Performance	35
6.5 Active Compensation Algorithm Performance	36
6.6 Environmental Impact	37
6.7 Discussion	39
7. Conclusion and Future Work	40
7.1 Summary of Achievements	40

7.2 Future Work	40
A. Appendices	45
Appendix A: List of elements per transceiver	45
Appendix B: CAD raport	46

1. List of Acronyms

FSO	Free space optics
RF	Radio frequency
ISM	Industrial, Scientific, Medical Band - Free to use
DDM	Digital Diagnostic Monitoring
OOK	On Off Keying
NRZ	Non-Return-To-Zero
RZ	Return-to-Zero
LOS	Line Of Sight
SFP	Small Form-factor Pluggable
EMA	Exponential Moving Average
BOM	Bill of Materials
WDM	Wavelength Division Multiplexing
COTS	Commercial Off-The-Shelf

2. Introduction

In a time of rapid global digitalization and a great increase in the need to process data, the demand for high throughput and reliable communication systems keeps growing. With the expansion of data rates, the need for security also increases. Classic Radio frequency (RF) systems, although widely used, might not satisfy the needs of all customers, due to the high traffic on unlicensed Industrial, Scientific, Medical Band - Free to use (ISM) bands and possible unauthorized listening.

However, there might be a situation where the fiber is also too expensive or simply too difficult to install.

Free space optics (FSO) communication might meet the requirements that RF lacks and gather the most demanding users. Using a precisely focused light beam, this solution can exceed RF solutions in terms of throughput, but also provides great security, by higher directivity and difficulty of hijacking the signal. Those characteristics make FSO systems a highly competitive answer to the need for private high throughput connections [28].

FSO is still being improved, despite having many advanced solutions, which are thoroughly described in academic papers, for example Deep Space Optical Communications (DSOC), that allows for over 2AU distance and over 200Mbs throughput transmission [11]. On the other hand there are also new solutions, that are dedicated for ground to ground applications, that utilize use of off-shelf components like SFP modules [26], [22], [1].

2.1 Background

The fundamental requirement of a well functioning free space optical link is maintaining uninterrupted direct line of sight, between transmitter and receiver. RF systems, on the other hand, diverge and bounce the obstacles or even pass through them and still can reach the destination. Ideally FSO systems, keep the transmitted beam collimated, which allows to achieve high power density and security of transmission at the expense of need for keeping precise alignment. In practical deployments, it is a challenging task because of environmental factors such as vibrations, thermal expansion, or atmospheric conditions. That is why

modern FSO systems are required to actively compensate movements of laser beam and its position.

2.2 Motivation

Although FSO systems are well described in scientific papers, developing such devices is a quite complex engineering task. The main motivation for taking this topic is the need to create an affordable and functional proof of concept transceiver and establish a high throughput link. Completing such connection is challenging because it requires knowledge of multiple fields: designing optical system, precise mechanics, electronics, and low level embedded system programming with real time requirements.

2.3 Thesis Objectives

The goal of this thesis is to design and develop a stable, high-speed FSO Ethernet link with active misalignment compensation. This will be achieved by using readily available SFP modules and a custom-built mechanical system for beam steering, controlled by a microcontroller that reads the Digital Diagnostic Monitoring (DDM) data from the SFP module to maintain optimal alignment. The implemented link will be used as a platform to analyze practical issues induced by environmental factors.

Ultimately, this thesis demonstrates that the integration of Commercial Off-The-Shelf (COTS) electronic components with custom 3D-printed mechanics provides a viable and highly adaptable framework for developing operational free-space optical (FSO) communication links.

2.4 Scope and Structure

This thesis focuses on the design, implementation, and evaluation of an active beam steering and alignment system for FSO communication. The scope describes the development of hardware and software solutions capable of sustaining and maintaining an optimal optical link. It explains the usage of SFP modules and the use of their built in DDM technology to extract real-time received power data. The practical work includes the mechanical kinematic stage and optical design of the transceiver, the development of a custom controller and the programming of necessary algorithms, such as spiral search and power alignment. Furthermore,

the scope includes experimental verification of the proposed system to evaluate its performance.

3. Theoretical Background and State of the Art

The need for communication and exchange of information is an important factor for human species for thousands of years, accelerating the growth of civilization. In the span of the centuries, methods of exchanging data have evolved from simple visual and sound signals, to first telegraphic and radio systems. The main breakthrough happened in the XX and XXI centuries, because of rapid digitalization, thanks to which time and distance stopped to be a barrier. Today, a fundamental medium on top of which our information exchange systems are built is optical fiber that allows to send encoded information in form of light into the cable. This chapter will introduce the essence of this technology and discuss in detail the principles of operation of fiber optic systems and how it can be applied in FSO communication.

3.1 Current Research Directions

Today, FSO transmission is being developed in couple directions, but most exciting are space applications. There are studies on optical inter satellite links, which allow for far higher throughput at lower power than RF and especially pronounced security [14]. There are also studies on modeling the atmospheric channel in ground to satellite communication, which are about to help in the future to implement such optical links [10].

3.2 Principles of FSO Communication

Generally, a FSO link is a set of two components: a transmitter and a receiver. The transmitter sends data signals free space using modulated optical radiation. It consists of radiating element, which is usually a laser diode, a laser modulator, and a set of optics responsible for creating a beam [20].

Modulation is essential in an optical transmitter. It allows data to be embedded into a light beam that then travels a significant distance to the receiver. The easiest way of modulating an optical signal is using direct modulation - changing

an injection current of a laser diode, which affects the output optical power. In this way, on-off binary modulation can be achieved, more widely known as On Off Keying (OOK) [13]. OOK simply turns on the light source to transmit a logic "one" and turns it off to transmit a "zero". There are two types of OOK which are Non-Return-To-Zero (NRZ) and Return-to-Zero (RZ). RZ version of the OOK simply encodes logic "one" as high amplitude followed by low amplitude, where NRZ does not return to low value, as shown in the figure 3.1. RZ has the advantage of higher sensitivity, but both modulations might lead to clock desynchronization during multiple zeros transmitted in a row [12]. OOK is proven to be working in FSO systems as it is used in other implementations [29].

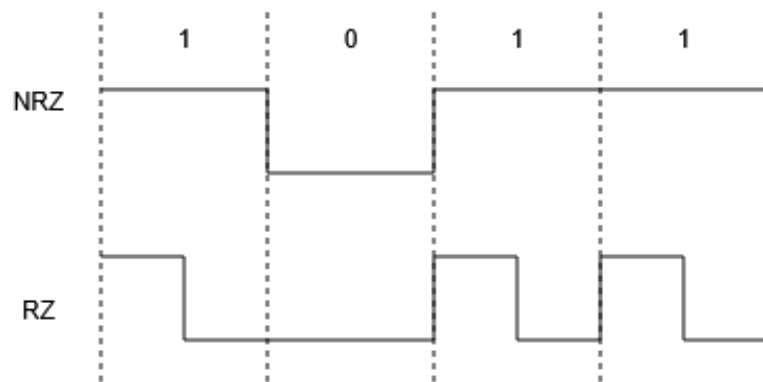


Fig. 3.1: OOK modulations

In order to increase the throughput of the network, Wavelength Division Multiplexing (WDM) is used. Instead of using one wavelength of the light to send data, multiple wavelengths are sent in the same optical channel as shown in figure 3.2. Different colors of the light do not affect each other and thanks to that properties, the overall throughput can be multiplied up to tens of Terabits [5].

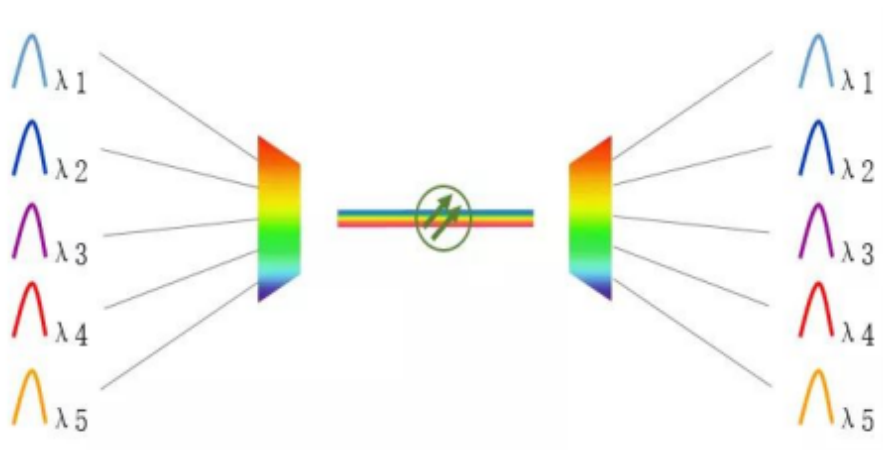


Fig. 3.2: WDM operation diagram. Source: [6]

Depending on how dense the wavelength channels are "packed" in the spectrum, this technology is called CWDM or DWDM as shown in figure 3.3.

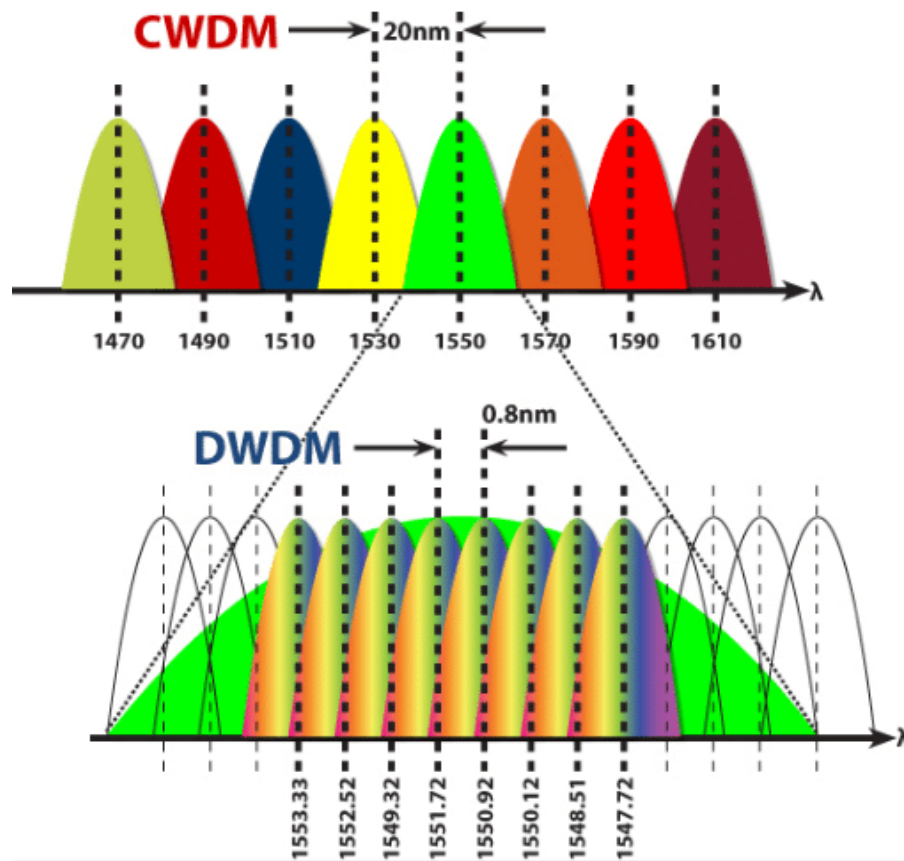


Fig. 3.3: CWDM and DWDM. Source: [7]

The modulated and multiplexed light beam now enters an optical system that often consists of a telescope or a collimator, which makes the light beam capable of traveling long distances. This signal needs to travel through an open space to the receiver. Line of sight between the transmitter and receiver is required. There are many factors that influence the path of the light, some directly like weather conditions and others indirectly such as vibrations, wind, or thermal expansion. Taking these into account, tracking systems must be applied [8], [16].

When the beam reaches its destination, it enters the optic systems, which focuses it at the photodiode, which converts light into an electrical signal. Then the signal is demodulated and the data is received at the other end of the link.

3.3 Impact of envirometnal factors on light propagation

The design and implementation of FSO systems require a solid background on the environmental factors that impact light propagation. Key examples include [21]:

- Physical obstructions: Temporary block of the Line Of Sight (LOS) can happen when objects, such as birds or debris, pass through the beam path,
- Scattering: Light signal can scatter when passes through water droplets. Fog, in particular, is the most significant limiting factor for transmission [17],
- Atmospheric turbulence: Temperature changes in the air create pockets with different refractive indices. Passing through these regions can cause the beam to change it direction,
- Atmospheric attenuation: Signal power is reduced over distance over a constant 0.225dBm/km [2],
- Physical misalignment: Temperature changes can cause thermal expansion or contraction in the mounting structures, leading to misalignment.

3.4 SFP Modules and Digital Diagnostic Monitoring (DDM)

The Small Form-factor Pluggable (SFP) modules are standard piece of equipment that can be bought at an affordable price. These are compact transceivers that are used in telecommunications, which translates differential electrical signal into the modulated optical signal.

Such modules inside are shown in the figure 3.4. The main elements are Laser Diode transmitter optical sub-assembly (TOSA) - used for transmitting data, and the receiving part uses receiver optical sub-assembly (ROSA) with PIN diode and transimpedance amplifier. In bidirectional one channel SFP both of those elements are replaced with BOSA (Bi-Directional Optical Sub-Assembly), which is combination of both. In addition to those main optical elements, electrical elements are also included in the PCBA, an amplifier used to boost the power of the received signal, Laser Diode Driver which directly modulates the laser diode. On the other side of the PCB, that can't be seen on the following picture is Digital Diagnostic IC for DDM implementation.

It is intended to be used in fiber cables that extend tens or even hundreds of kilometers. SFP modules have various type of connectors, like RJ45 for electrical, LC and SC for optical. As written before, these modules are transceivers, allowing

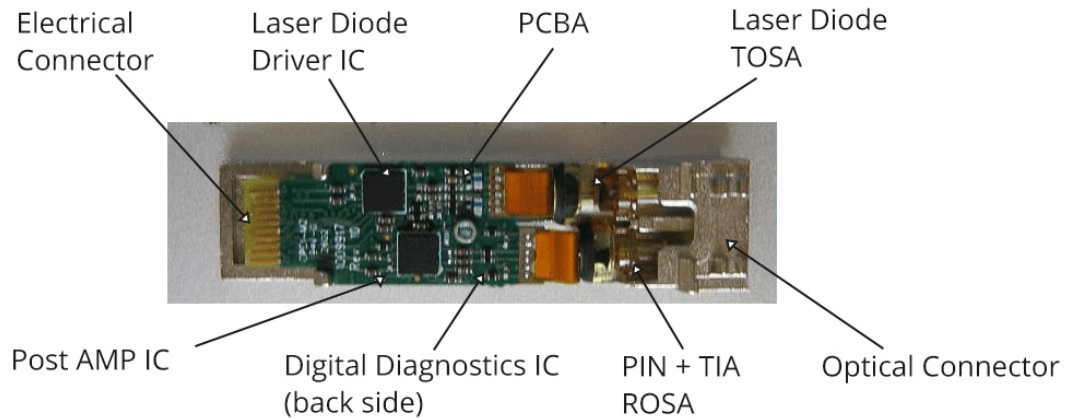


Fig. 3.4: Dual channel SFP, Source: [18]

for full duplex communication. It is realized by two physical optic channels or one optic channel using two different wavelengths (eg. 1550nm Rx/1330nm Tx).

Typically SFP uses wavelengths of telecommunication windows, which are 1st window - 850nm, 2nd window - 1310nm and 3rd window 1550nm. These wavelengths are perfect for fiber communication because of the low transmission loss in the fibers [3].

The SFP modules are equipped with DDM, which is management technology that allows users to monitor in real time a few transmission parameters. Those parameters are: output optical power, received power, temperature, laser current, and transceiver supply voltage.

The mentioned characteristics make an SFP modules a great candidate for the base element of the FSO transceiver. It drives the laser, handles the modulation, and also implement the receiving path. Different connectors and wavelengths configurations allow for different designs of the transceiver. DDM allows for real time data harvesting, which is essential for a tracking system.

3.5 Existing Beam Steering Techniques

FSO link during its existence might destabilize, which can lead to loss of received power, reduction of throughput, and in the worst case total loss of connection. To limit the impact of environmental factors that destroy the connection, providers of FSO links are implementing beam steering into their links. On top of that alignment of long range link, without any beam steering is a tedious task, as it would require perfect alignment of two nodes. Typical application of FSO link environment conditions are changing, such transceivers are usually placed outdoors and are affected by changing thermal expansion of the materials and

structures they are placed on [24]. In such a case, the perfect kinematic mount, which is perfectly aligned, can change an angle a little bit which can lead to major shifts over the deployment distance. At first glance it seems to be unnecessary, because in lab conditions the link might work correctly, but the deployment of a long range link is a hard reality check.

The easiest way to provide beam steering into the design is to create a mechanical structure capable of rotating the whole transceiver. Using such mechanism can provide big coverage and it is a great solution when transceivers are placed on mobile platforms, for example UAV drone.

The next possible solution is to implement translation into the optical system. Movement of the lens or the mirror used to launch the beam allows for change of angle in which the beam exits the transceiver.

Lastly, the translation of optical element could be swapped with a tilt of the lens or the mirror, but tilting might introduce an increased level of difficulty, of understanding and modeling such movement.

4. System Architecture and Hardware Design

Developing an FSO link involves significant engineering challenges, requiring the precise integration of mechanical, optical, and electronic systems to ensure reliable, high-throughput transmission. This chapter details the overall architecture and mechanical solutions implemented in the designed devices.

First, a general overview of the transceiver and the link is provided, including the selected components and the final system parameters. Next, the mechanical and optical designs of the device are presented, followed by a description of the controller's electronic circuitry. Finally, the chapter outlines the software architecture and explains the design choices.

4.1 Transceiver and Link Overview

The architecture of the transceiver was carefully chosen to be realistic in completing in a given time frame and also competitive on the market. Another constraint was the affordability and availability of the elements on the market. The main elements of it are:

- gigabit SFP media converter,
- full duplex, 1.25GB/s, 80km, WDM, SFP 1330/1550 (rx/tx or tx/rx depending on the unit) SM SC connector providing one optical path with DDM,
- 3d mechanical movement allowing for SFP translation,
- Nema 17 1.8 deg per step
- plano convex lens 50mm diameter, 85mm focal length,
- electrical control system for driving the motors and tapping into SFP's DDM,
- esp32c6 for low throughput RF transmission.

This list of elements gives competitive parameters to the FSO transceiver and keeps the cost relatively low. The Bill of Materials (BOM) with costs is presented in Appendix A.1. The final cost for single device with such elements is 550 PLN, which is 1100 PLN per link capable of uninterrupted 1Gbs transmission at tens of meters.

In figure 4.1, the proposed system diagram is presented. There is a simplified block scheme of transceiver shown and the link itself.

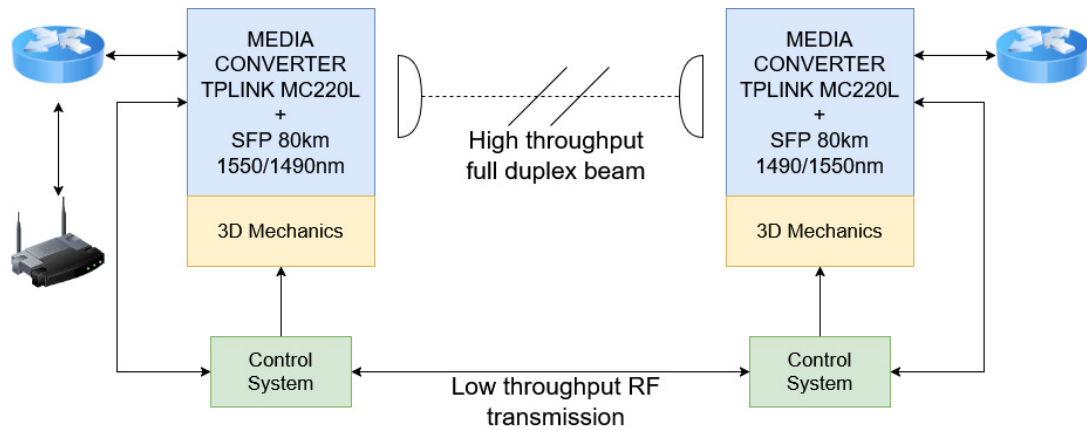


Fig. 4.1: Transceiver and System diagram

4.2 Mechanical and Optical Design

The crucial part of the transceiver is the optical and mechanical design. In this particular case it is embedded together and movement provided by mechanics changes the position of the light source to the lens. It changes the angle in which the light exists the transceiver, allowing beam steering.

In case, when light source is placed in the focal length - the light beam will be collimated. Approximately if we move it in parallel to the lens, the beam will be almost collimated and will exit the lens at the angle. If there is a second light beam coming into the transceiver at the same angle as the exiting beam, it will focus in the same place as the light source. Due to this characteristic, this beam steering technique can work, allowing transceivers to communicate when they are shifting from ideal alignment, as shown in 4.2.

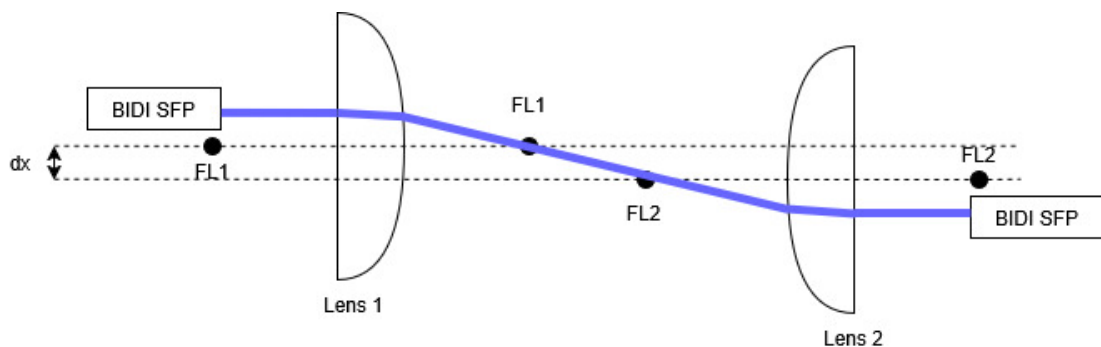


Fig. 4.2: System of 2 transceivers

To achieve such beam steering, proper mechanical system has been created that allows for 3 axis movement of the SFP module. First axis, let's call it x is perpendicular to the lens and translation alongside this axis change the distance

of SFP to the lens. Another two axes, y and z are parallel to the lens and moving the source in those direction steers the beam, as shown in the figure 4.3.

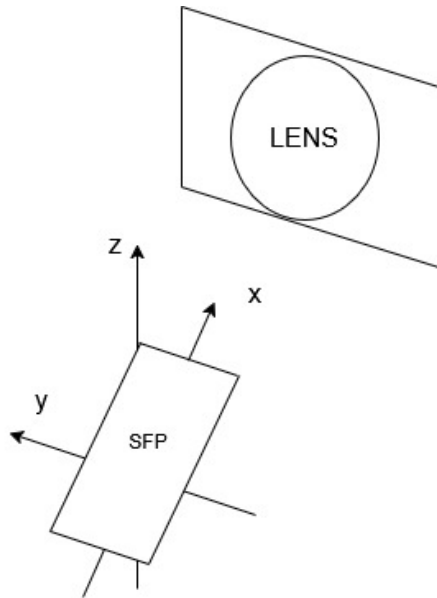


Fig. 4.3: Translations of transceiver

The mechanical structure allowing movements alongside those axes has been based on lead screws and anti-backlash nut's rotated by stepper motors. The used screw has 1mm pitch, which means that one full rotation of the screw will move the element by 1mm. Besides the lead screw, nuts and bearings, the rest of the mechanics was design for 3D printing. Although 3D printing does not give best results of the mechanical structure (Low stiffness, thermal properties and precision), it is great for rapid development and prototyping.

The 3D printed structure also includes guides that the elements move on. The shape of those guides on all axes was a trapezoidal shape, as can be seen in the figure 4.4. On the figure 4.5 there is shown a hole which fits the guide rails. There are circles placed in the corners to eliminate the jamming of the sharp edges.

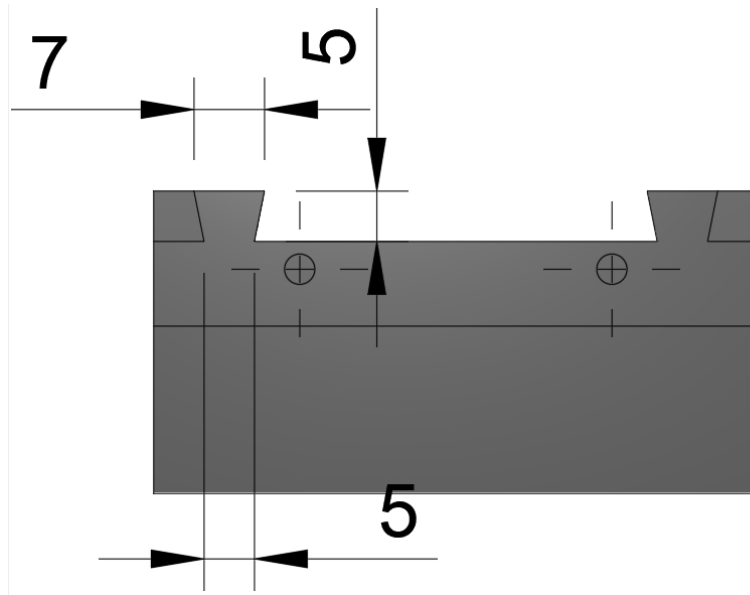


Fig. 4.4: Guide rail

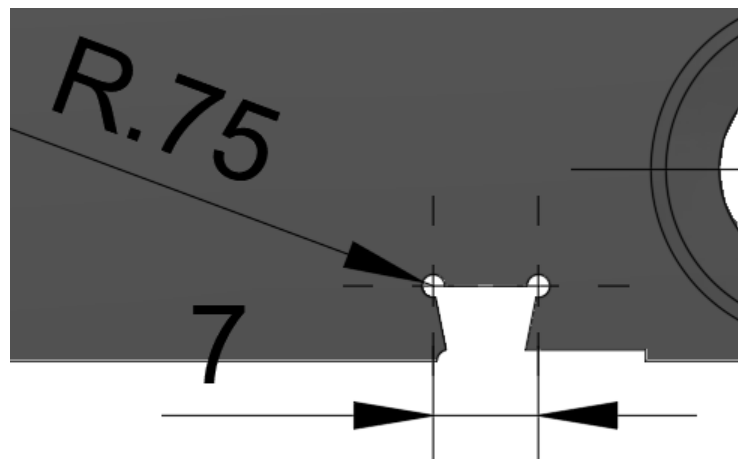


Fig. 4.5: Guide hole

Mechanical movement was limited to a 10mm x 10mm range of movement parallel to the lens by end switches. Such a range gives quite a large range of motion, so it made a low range tests a bit easier. The final CAD model of the mechanical structure is provided in Appendix A.2 and manufactured transceiver is shown in the following figures 4.6, 4.7, 4.8, 4.9.

In figure 4.6, the frontal photography of the transceiver is shown. From here, the bearing for the X axis (Movement perpendicular to the lens) is visible, labeled "1". On the right side there are "Iron sights" aligned, marked "2", used to aim at the other transceiver.

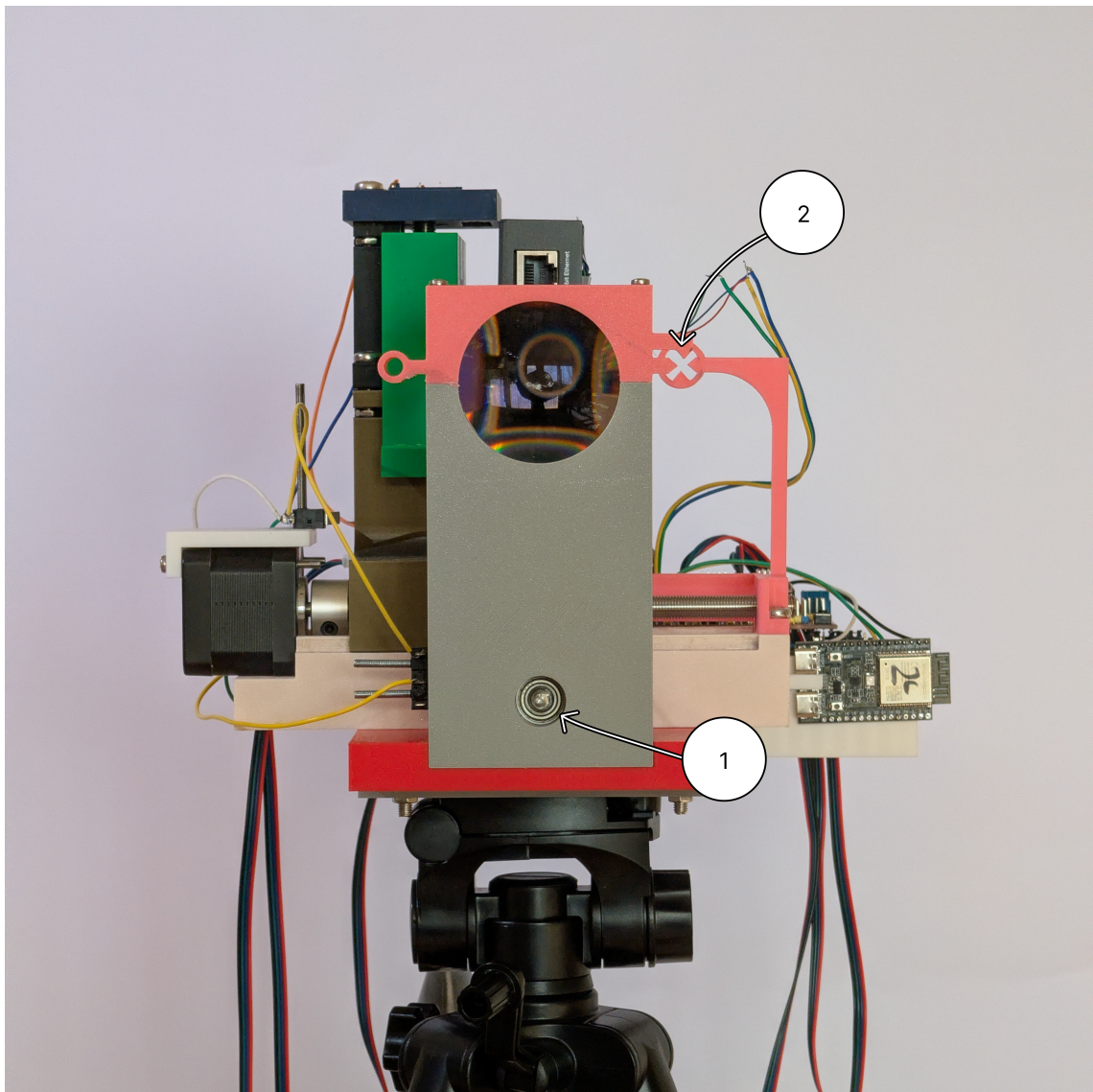


Fig. 4.6: Transceiver photo no.1

In figure 4.7 the front right photo of the transceiver is shown. As can be seen, the lens is placed in a housing and is held in place by two screws.

By "1" the one of the Y axis end switches is placed, it is connected to the bearing mount.

This photograph nicely showcased how nuts, marked as "2", are placed in the 3D printed structure - they are pushed into and sit in the place nicely, similar to the bearings marked as "3".

On the right in the white box, marked as "4", the controller is placed with convenient placement of ESP32C6 dev module, allowing for sufficient range of RF transmission [23].

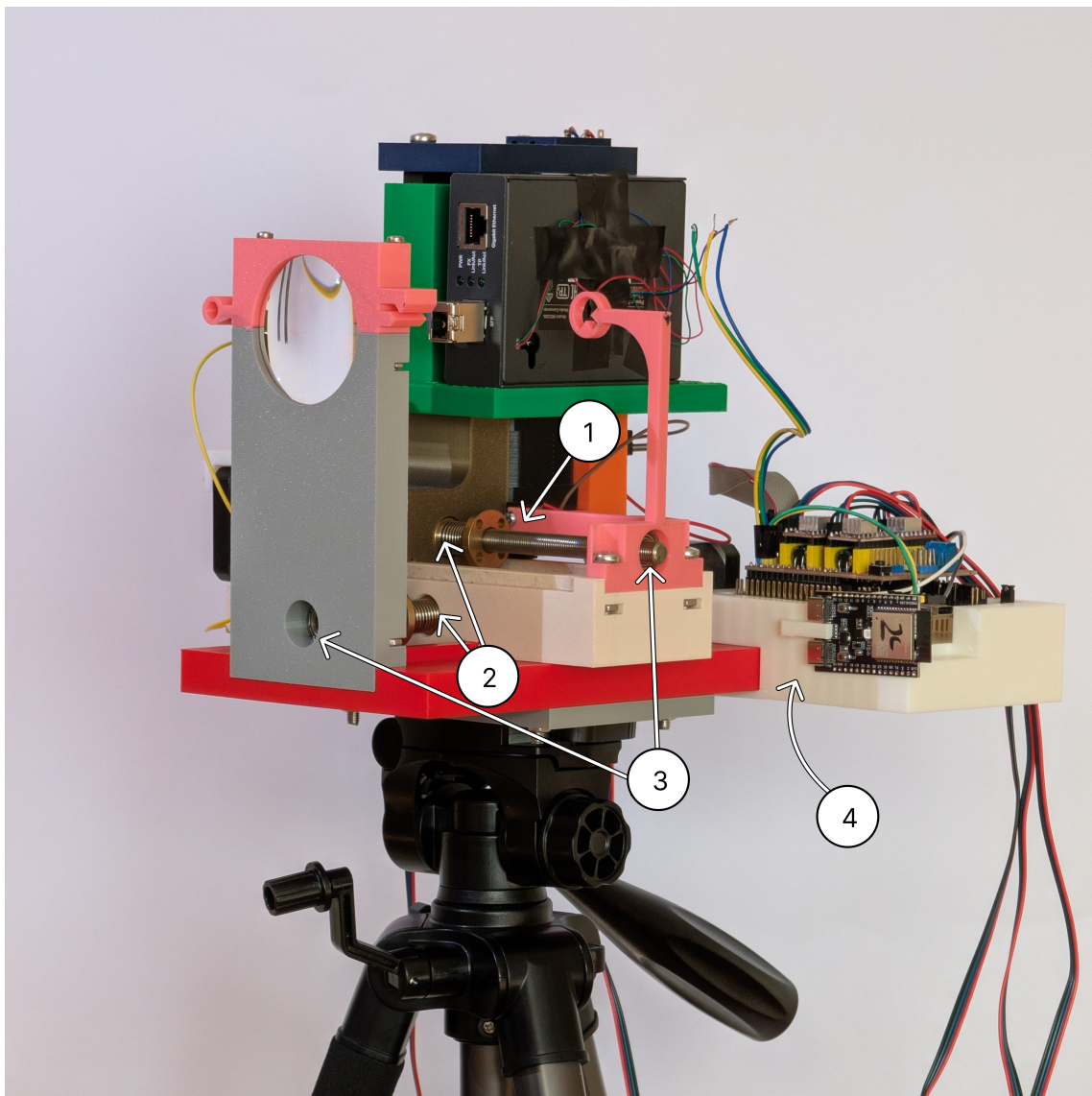


Fig. 4.7: Transceiver photo no.2

In the figure 4.8 is a picture of the transceiver from the other side. Marked as "1" is one of the Y axis end switches, which is mounted to the stepper motor itself. Marked as "2" is the only end switch for the X axis, which is crucial for resetting the position to the known state. Also, it can be noticed that the whole structure is placed on the tripod, by custom made mount.

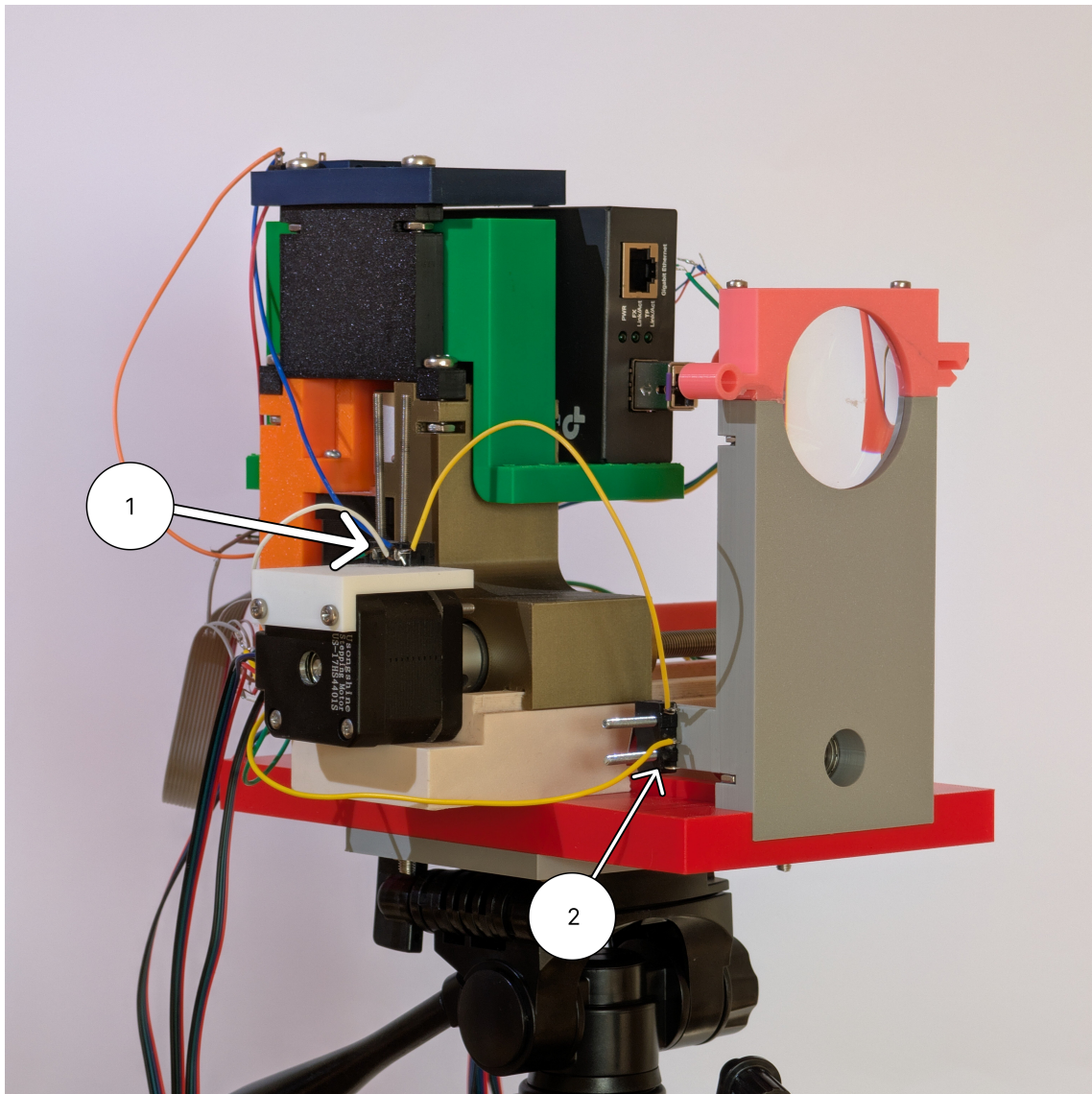


Fig. 4.8: Transceiver photo no.3

Figure 4.9, is a photograph of transceiver from the back and it shows the end switches of the Z axis marked as "1" in the picture.

It also shows three stepper motors for the X, Y, and Z axes, labeled "2", "3", and "4", respectively. These motors are mounted using screws into the design. On the left side you can notice how the white box with controller is pushed into the design next to the motor.

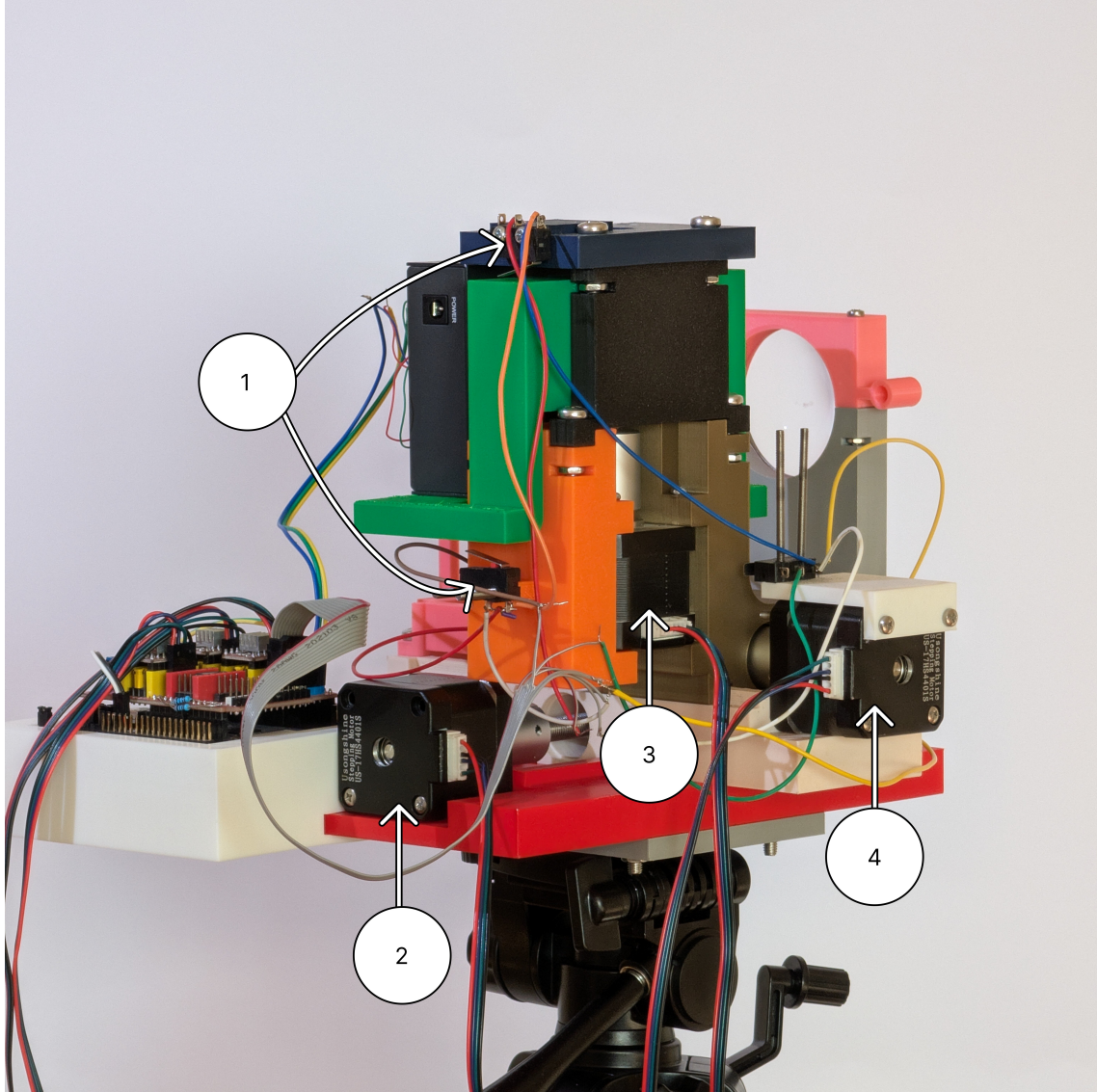


Fig. 4.9: Transceiver photo no.4

4.3 Controller Design

A controller is an important part of the project, as it is running the whole transceiver. It consists of couple elements connected by GPIO's or communication protocols.

- the main MCU of the controller is the STM32F1 microcontroller,
- it's outputs are connected to the motor controllers A4988, which are configured for 4 x microstepping,
- inputs are used to connect the end switches connected in series, so if any of the switches of the given axes are closed, then it goes to one pin,
- I2C bus is used to connect the microcontroller to the SFP module, to hijack out the DDM data,
- There is also a UART connection to the ESP32 microcontroller with a PCB antenna serving as an RF transceiver for communication in the link.

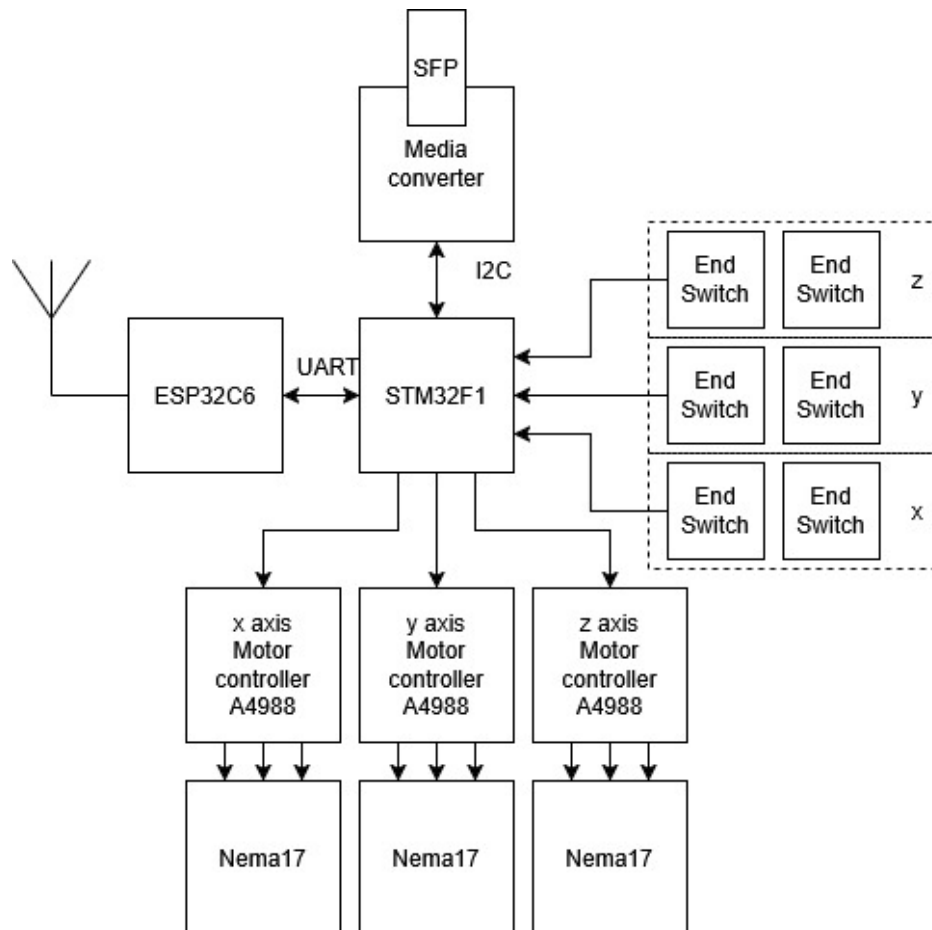


Fig. 4.10: Block scheme of the controller

The firmware, which is run by the STM32 microcontroller, is a Non-Preemptive, Polling-Driven Single-Tasking Architecture, which calls multiple Finite State Machines (FSM). The tasks it runs are:

- Movement Handler - it drives the motors and checks the end switches,
- Spiral Search FSM - controls the spiral search for setting up a new link,
- Homing FSM - controls the homing procedure to position the SFP in calibrated position,
- Serial Comms - checks the UART communication, from here FSM's can be toggled from sleep to run,
- DDM Readout - reads the DDM data with given period,
- Optimize FSM - algorithm that tries to optimize the received power for given distance from the lens,
- Active Compensation FSM - algorithm that keeps link at the optimal power.

It is shown graphically in figure 4.11.

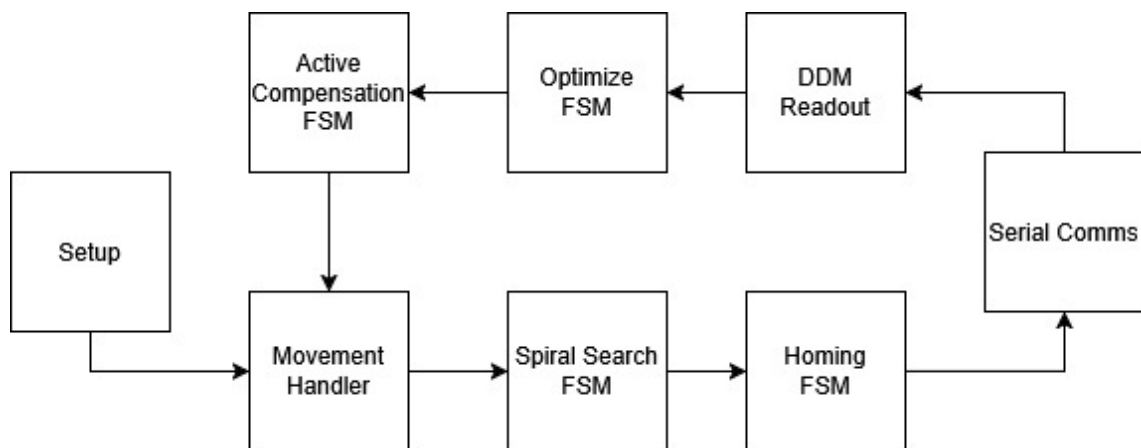


Fig. 4.11: STM32 firmware diagram

ESP32 has its own firmware that runs freeRTOS and is responsible for information exchange with between two transceivers and lets operator connect to a link via the UART. Three concurrent tasks are run:

- PC Serial Comms - is a task that polls for communication from PC and forwards it to the one of the transceivers,
- STM32 Serial Comms is a task that polls for messages coming from STM32 and forwards them to the PC or second transceiver,
- ESP-NOW Message Handler is a task that polls for messages coming from the other transceiver and redirects it to the PC or the transceiver it is connected to by a UART.

It is shown graphically in figure 4.12.

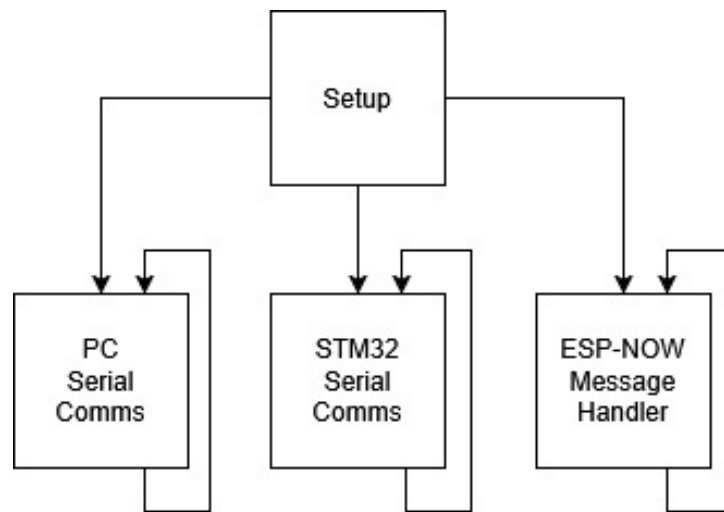


Fig. 4.12: ESP32 firmware diagram

5. Active Control Algorithm

This chapter explains how the built platform is controlled, starting from how data is collected and then how this information is used to set up the link and keep it running.

5.1 DDM Data Extraction

As mentioned before, most of the modern SFP's have DDM implemented, which can be accessed by connecting to SFP's I2C pins, or using media converter functionalities. Unfortunately in this case, used media converter doesn't provide DDM data interface, so controller must be soldered into SFP pins.

SFP pinout is provided in the figure 5.1. SDA is at pin 6 and SCL is at pin 5.

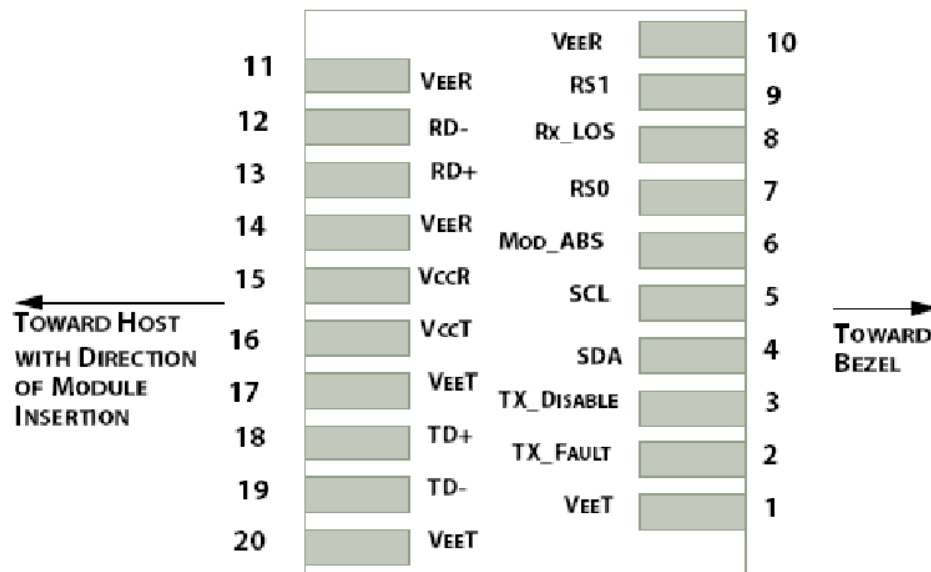


Fig. 5.1: Pinout of the sfp. Source: [25]

There is a lot of different information about the SFP module that can be accessed via this I2C, but the most interesting part is the "Real Time Diagnostic and Control Registers" part. It is a set of variables that are normally used to diagnose the module and link health.

Specific addresses can be seen in table 5.1.

A2h	Bit	Name	Description
Converted analog values. Calibrated 16-bit data.			
96	All	Temperature MSB	Internally measured module temperature
97	All	Temperature LSB	
98	All	Vcc MSB	Internally measured supply voltage in transceiver
99	All	Vcc LSB	
100	All	TX Bias MSB	Internally measured TX Bias Current
101	All	TX Bias LSB	
102	All	TX Power MSB	Measured TX output power
103	All	TX Power LSB	
104	All	RX Power MSB	Measured RX input power
105	All	RX Power LSB	
⋮			

Tab. 5.1: Real Time Diagnostic and Control Registers [Address A2h, Bytes 96-111] [27]

In addition to this table, there is also information on how to interpret these values, according to SFF8472:

"Measured RX received optical power in mW. Value can represent either average received power or OMA depending upon how bit 3 of byte 92 (A0h) is set. Represented as a 16-bit unsigned integer with the power defined as the full 16-bit value (0-65535) with LSB equal to $0.1 \mu W$, yielding a total range of 0 to 6.5535 mW (-40 to +8.2 dBm) [27]."

There is no much information on the refresh rate of these data in this specification document, so that had to be checked empirically. In case of the SFP modules that were used in this transceiver, the data update period was $t = 15ms$, which corresponds to $f = \frac{1}{t} = \frac{1}{15 \cdot 10^{-3}} = 66.67Hz$.

Having this information, it can be concluded that with this data rate, the transceiver will be able to compensate for low frequency changes.

5.2 Connection Seeking Algorithm (Spiral Search)

After placing the transceivers so that they are aimed at each other, it is demanding task to achieve perfect alignment. It is because of imperfections of the mechanical design, or even media converter itself, the SFP module can be a little bit rotated. It can be calibrated coarsely, but in the end a search algorithm had to be developed.

The algorithm applied is the Square Spiral Search Pattern Method. Due to

high directivity and an imperfect homing sequence, the spatial search algorithm must be applied to connect both transceivers. The flow chart has been presented in the figure 5.2. The algorithm has an asynchronous stop condition. The lower the dx the denser the search. That dependency allows for using the search in different scenarios, like in case the connection was lost dense version will be appropriate. During link set up, rather coarse version is suggested, because the field of search is much bigger.

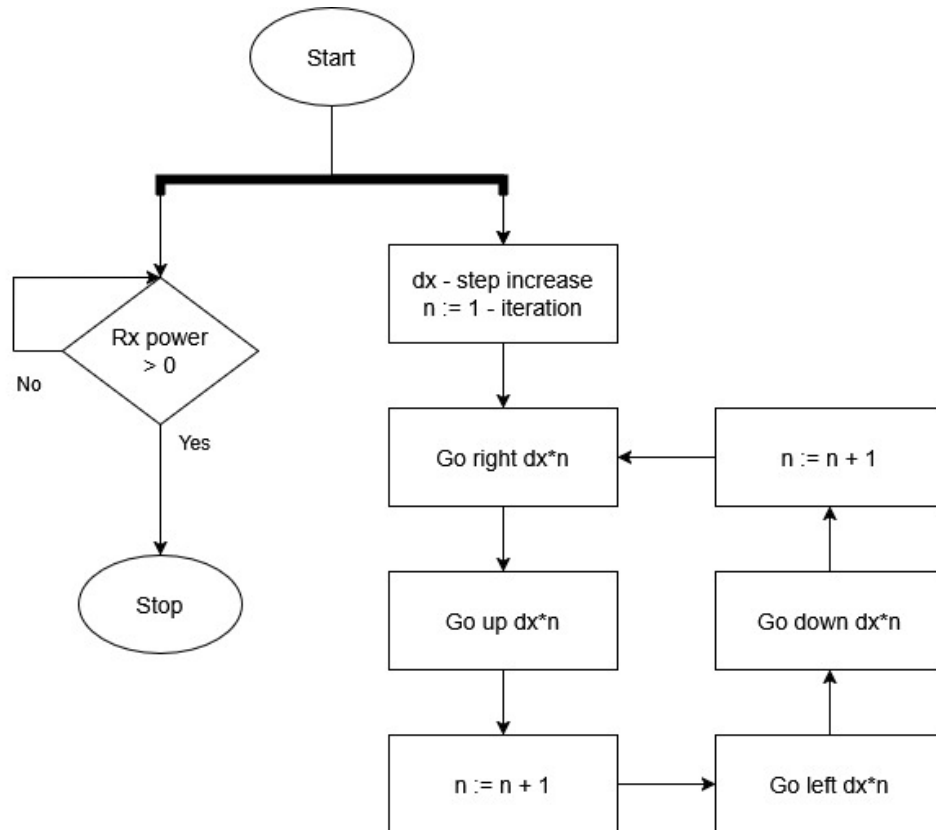


Fig. 5.2: Spiral search flow chart

5.3 Power alignment algorithm

After running a spiral search, the received power is at a low, nonoptimal power level. The purpose of Optimizing Algorithm is to find a maximum received power point, by controlling the Y/Z axes. In this moment the continuous connection isn't required, since it's still running during link set up.

The proposed algorithm is a bounded linear search, the flow chart is shown in figure 5.3. It sweeps three times to ensure that it approaches the peak from both sides and in case it starts going in wrong direction. The change direction condition is limited to the quarter of current maxPow, to make the algorithm a than a full sweep. The 50 ms delay between moving and currentPow readout lets the

mechanical vibrations settle to make the readout undistorted. Firstly, such an algorithm should be applied to one axis and then another [4].

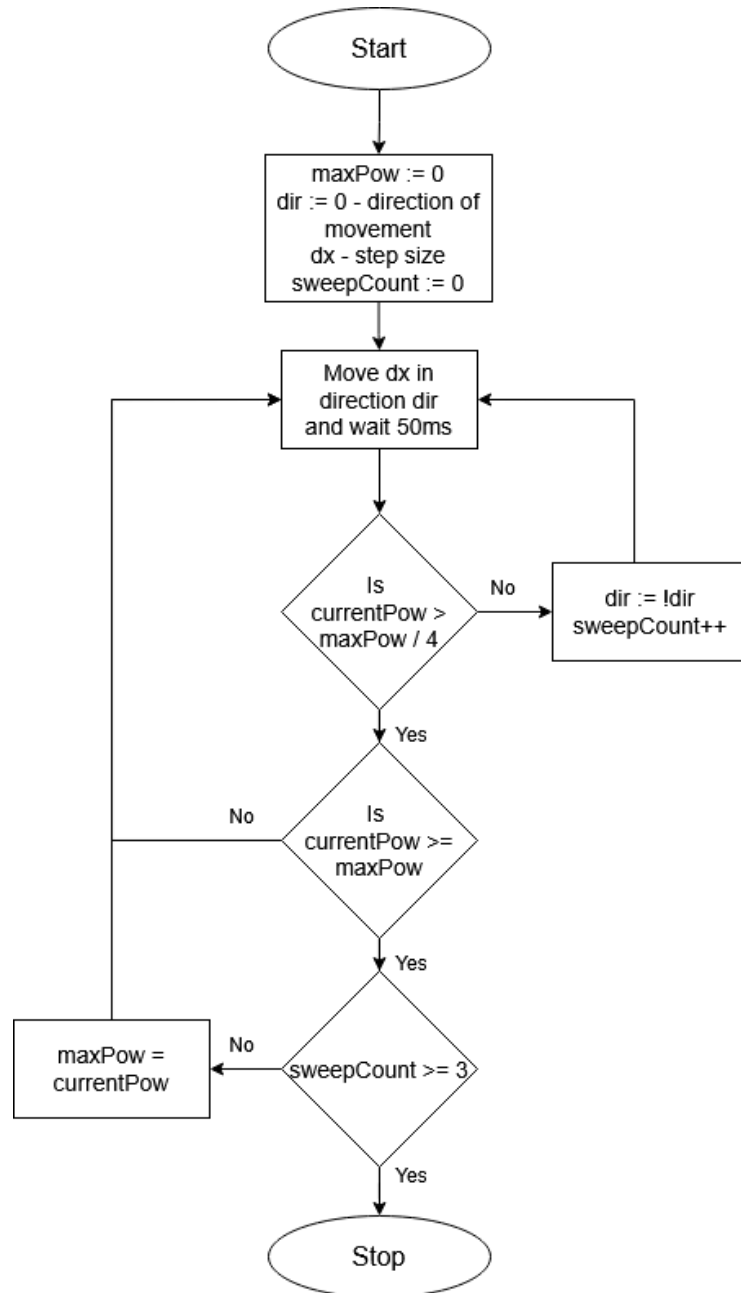


Fig. 5.3: Bounded Linear Search

5.4 Control Algorithm Design

When the link is already set up and used for data transfer, the continuous transmission is preferred, so an active compensation cannot use the same algorithm, which was used during link set up to maximize received power. As stated in

DDM section, refresh rate of received power in DDM is rather slow, the algorithm to actively compensate the changing value doesn't need to be really fast.

On top of that the power curve is rather simple with one maximum, since it is the algorithm that is applied in Y and Z axes (parallel to the lens), since X axis (perpendicular to the lens) should be already well align after link set up. The Control Algorithm is presented in the form of a flow chart in figure 5.4. This procedure differentiates from the previous algorithm by the execution time, since it does not scan broad range, rather observe how values changes in real time. It guarantees that final state is just a step away from maximum power. This algorithm should first be applied to one axis and then to another [9].

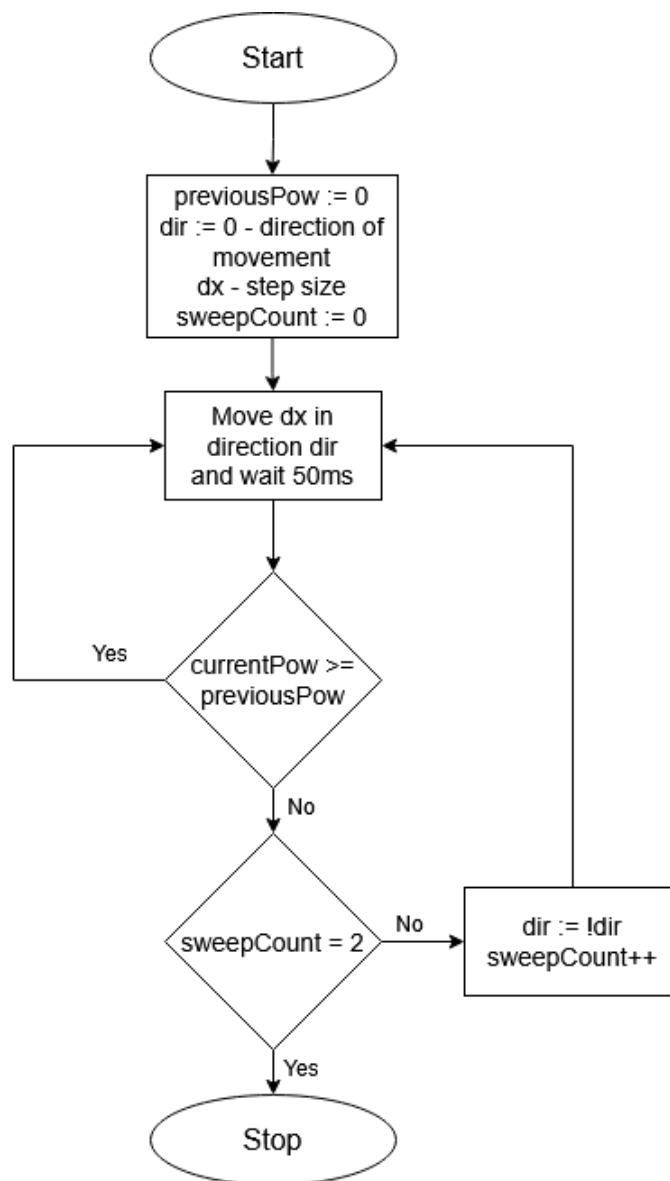


Fig. 5.4: Control algorithm flow chart

5.5 Misalignment Compensation

There are a few more nuances in the active compensation. Firstly, the reading from the sensor might fluctuate and can be affected by high frequency changes.

To get rid of unwanted readings, some sort of low pass filter must be applied. The LPF of choice in this case was Exponential Moving Average (EMA). It is an algorithm that does not require memory for saving the samples, in comparison to other moving averages like Simple Moving Average (SMA) or Weighted Moving Average (WMA) [19]. The equation for an EMA is:

$$y_i = \alpha * x_i + (1 - \alpha) * y_i \quad (5.1)$$

where y is the output, x is the input and α is a constant in the range $[0, 1]$.

Another point is that the link is established by two transceivers. Running algorithms without communication between 2 ends would lead to strange results and decalibration. To make it convenient, RF communication was introduced into the design.

The basic handshake is introduced, as can be seen in 5.5.

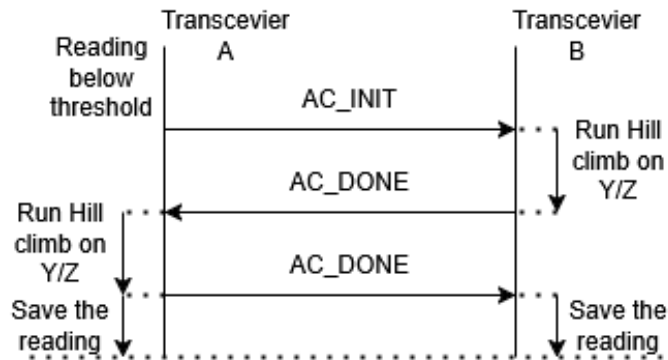


Fig. 5.5: Execution flow

Such a sequence should guaranty good execution of hill climb algorithms in both transceivers of a link. The order is that the transceiver initializing the compensation dispatches Control Algorithm on the second transceiver first ensures that no algorithm will be run when there is no working connection.

6. Experimental Setup and Measurements

In this chapter, experimental methodology will be explained. The setup of the test, scenarios in which link will be placed, and measurement metrics. Next, the measurements will be shown with proper analysis.

6.1 Experimental Setup

The transceiver itself has been equipped with a custom made tripod mount, to allow for simple placement without the need for time consuming installation.

The environment in which the link was tested was an outdoor free-space test. The transceivers were placed in two separate buildings and spaced 23 meters apart according to the rangefinder measurement, shown in figure 6.1.

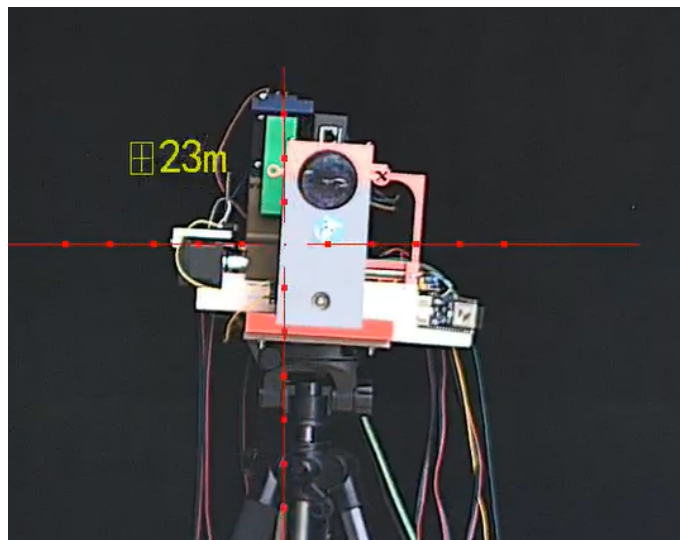


Fig. 6.1: Rangefinder measurement

Such an environment is ideal for testing a link, because it is real application of such link, where the beam passes through a free space and can be affected by weather conditions. The diagram is shown in the figure 6.2. Between buildings, rain and other factors can affect the beam.

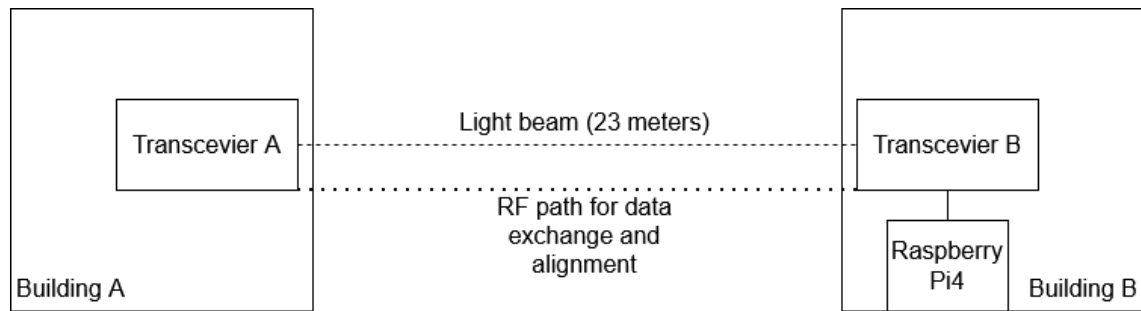


Fig. 6.2: Environment

In figure 6.3, a pair of transceivers are shown, These devices were used during a measurements. Black transceiver served as master, which was connected to the SBC computer - Raspberry Pi4, that allowed for remote collection of data.

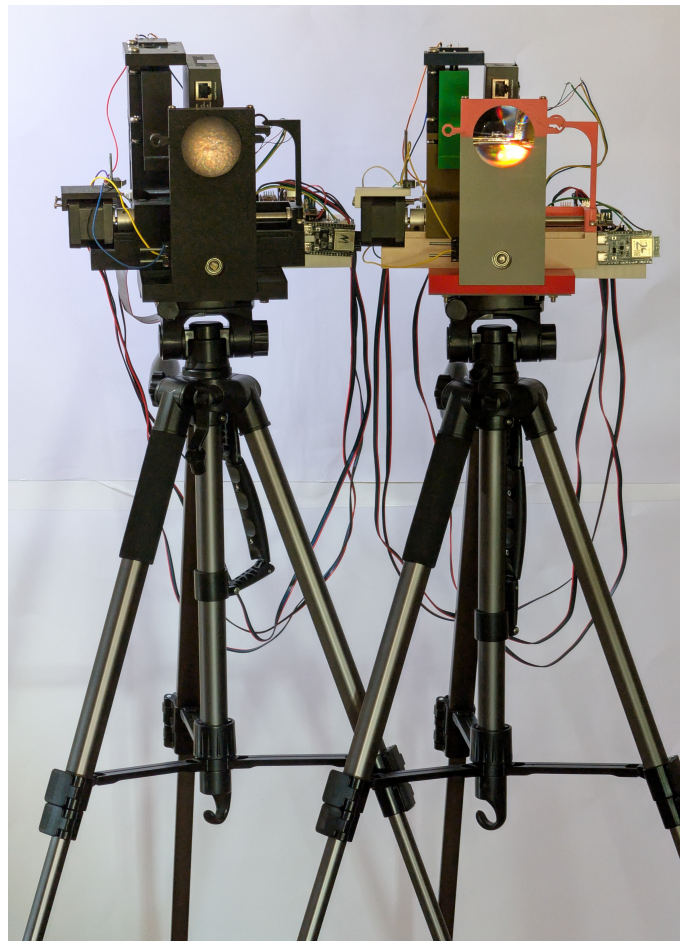


Fig. 6.3: Pair of transceivers, used in tests

6.2 Test Scenarios and Measurement metrics

The test scenarios were dictated by the weather conditions in this example. The conditions on the test day were consistent with weather conditions change, varying from clear to runny and windy.

The measurement metric that was collected during the tests was received power. It was collected by the Raspberry Pi4 computer connected to one of the devices. The other end measurement was provided by RF interface ESPNOW.

6.3 Link throughput measurement

In addition to measurements of compensation performance and weather influence, empiric tests were conducted on link bitrate in dependence to the power received.

It was carried out in an indoor environment, in a range of 10 meters and bitrate was tested with iperf3, which is a tool for active measurements of the maximum bandwidth on IP networks [15].

Received Power [μ W]	Download bitrate [Mb/s]	Upload bitrate [Mb/s]	Retransmissions
20	948	958	40
10	949	956	32
2	949	957	41
1.5	929	937	617
1.3	447	454	85

Tab. 6.1: Download and upload bitrate in relation to received power

6.4 Baseline Performance

The baseline for the research is the performance of the FSO link without active compensation turned on. Such measurements have been collected and are presented below in the figure 6.4. The blue line is a measurement from the master transceiver which is the one with Raspberry PI4 connected. The orange line represents the received power of the second transceiver.

In this test weather conditions were okay, without significant temperature change. The results tend to slowly fall down in this case. Over 600 seconds of running, received power decreased by almost 50uW, from 430uW in the case of Master transceiver. The slave starting performance was at 200uW and fall by 30uW.

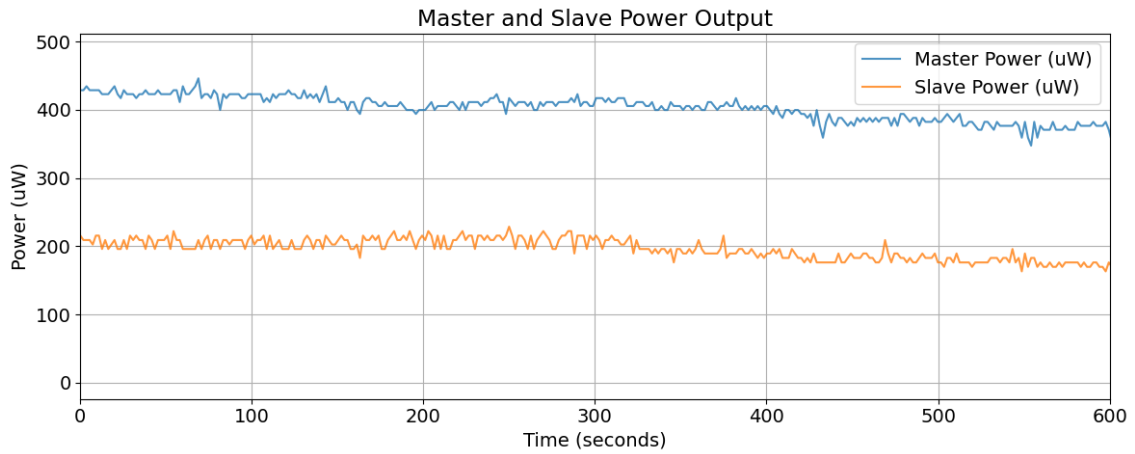


Fig. 6.4: Measurement without active compensation in good conditions

In another scenario, shown in the figure 6.5 without active compensation, the conditions had more influence on the results. In this case starting power measurements were 156uW for master and 35uW for a slave. During the test the laser beam must have drifted a bit due to environmental factors, lowering the power from peak point on master transceiver to 65uW which is a significant drop.

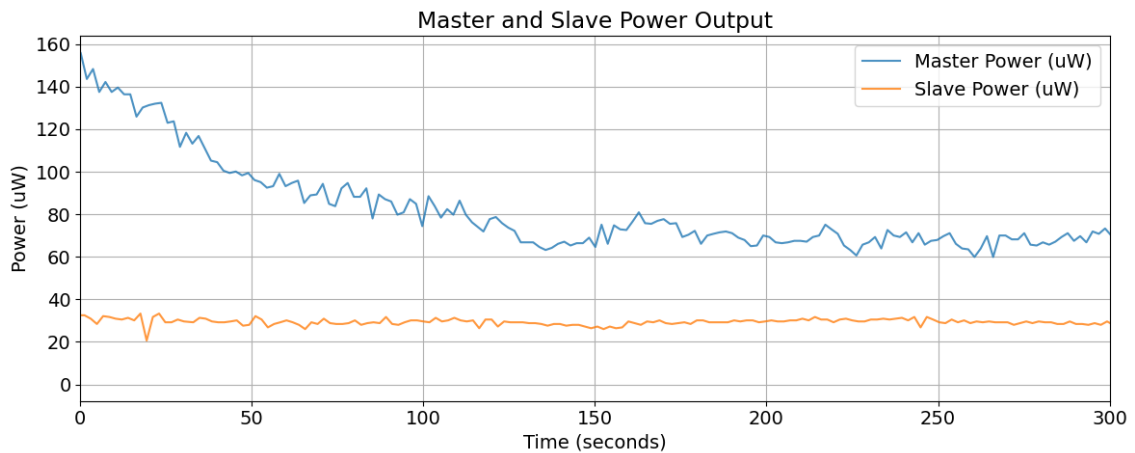


Fig. 6.5: Measurement without active compensation with power drift

6.5 Active Compensation Algorithm Performance

To compare with baseline performance, active compensation performance was also measured. In figure 6.6, the performance of both Master and Slave transceivers is visible. The blue line is a measurement from the Master transceiver, where the orange line is the slave. In this plot there are red vertical lines that indicate AC INIT event, which is the initialization of the active compensation sequence. After the sequence point of the new P_{max} is marked, which is used

to dispatch new compensation events if the power drops below $80\% * P_{max}$ or increases above $120\% * P_{max}$.

The factors that affected the transceiver were the weather conditions outside. Overall, the output power was improved throughout the measurement, by finding new maximum output power. Looking at the baseline performance, the active compensation works as intended and serves its purpose by not letting the link degrade.



Fig. 6.6: Measurement with active compensation

6.6 Environmental Impact

As mentioned in the baseline performance case during the measurement, it was raining at the end. In figure 6.7, the last 300 seconds are presented, when heavy rain was affecting link quality.

Throughout the measurement, there is strong variation visible at the moments of rain, even dropping the output power temporarily to 0uW, sometimes the beam was affected in the other way, spiking the received power of Master transceiver to 600uW.

This case shows how dense amounts of droplets of water might even block the connection, causing loss of packets, which decrease the throughput.

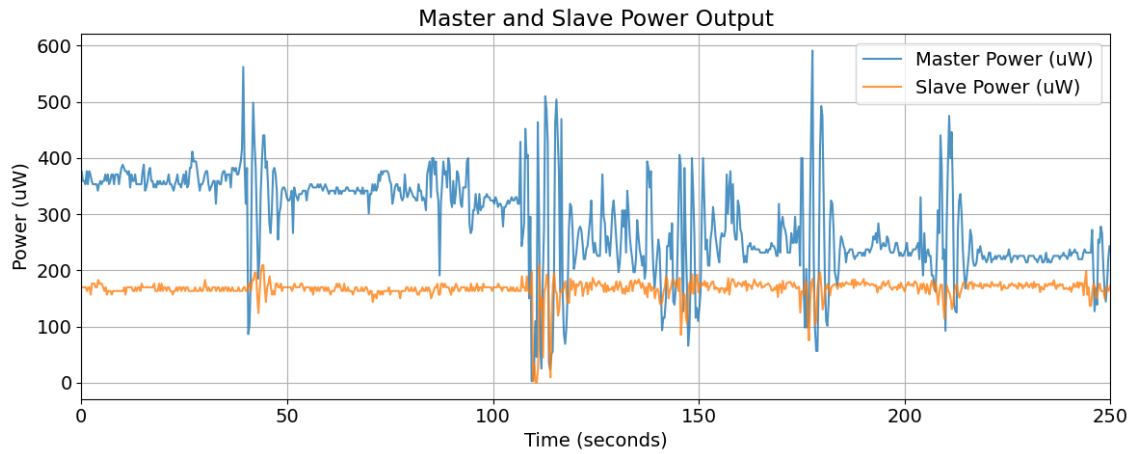


Fig. 6.7: Heavy rain effects on received power

There was another measurement on a rainy day. It is a 3700 second measurement starting from relatively low 50uW received power in Master and 30uW received power in Slave. The measurements in this time were affected by heavy rain and strong wind. The results show that Master received power changed from 50 to even above 200uW, and those results showcased moments, where received power dropped down to 0uW and at the end the link fully destabilized.

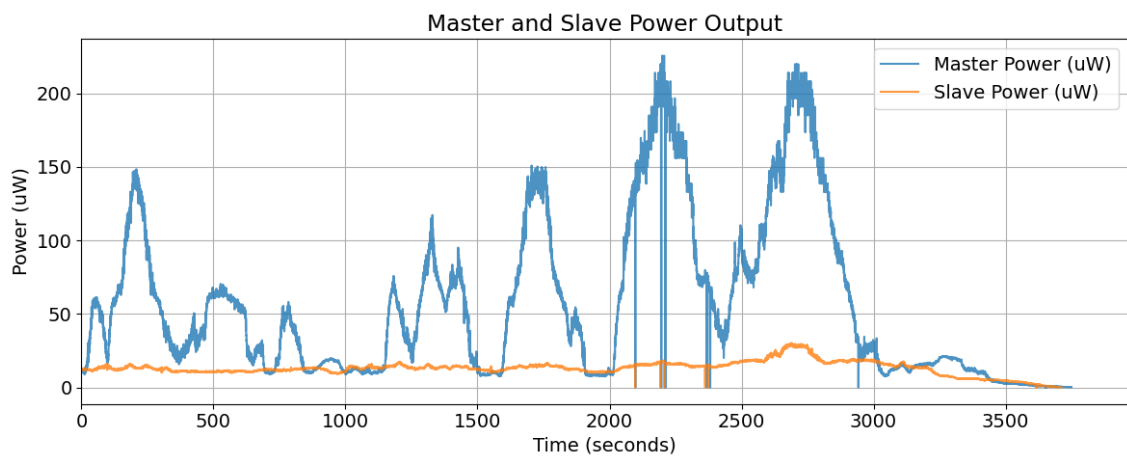


Fig. 6.8: Highly changing environment with rain and wind

6.7 Discussion

In the baseline performance, two cases were observed, conditions were received power stayed stable and one with significant drop of received power. Although power metrics stayed in reasonable range allowing for full throughput, in longer range links the drift could destabilize the link.

The results of Active Compensation in comparison to baseline performance show positive results - the received power metric was controlled at kept at good level. There are some nuances that can be taken into account while improving the overall performance. Currently, the transceiver is optimizing the received power of itself, but refined version would be optimizing the sum of both received powers. The biggest bottleneck of the system is the DDM polling data frequency, 15ms makes a transceiver appropriate for not moving applications. If there was an intention to apply this communication in drone communication to vehicle communication, the update would be too slow and would require other input data to base the compensation. Also, such application would require a much higher range of motion. The current mechanics could be a nice base for "fine" steering in such a system, where "coarse" steering would be a rotation of the base of transceiver.

Overall, the 3D printed mechanics and the steppers motors give nice stability for "last mile" applications, allowing for high throughput secure connections where it is impossible or too expensive to lay fiber. This is also a pretty good base, which could be extended for more advanced application, which would run real time tracking, but that would require to swap current SFP module for custom electronics, allowing for faster data harvest.

7. Conclusion and Future Work

This chapter will conclude the work described in this thesis and speculate about possible improvements to the device.

7.1 Summary of Achievements

The project was carried out successfully according to the objective of the thesis. A stable and high throughput optical link was design and produced, capable of 1Gbs transmission over tens of meters. It was achieved by using readily available and standardized components. The prototype devices produced were used as a platform to analyze the influence of weather conditions on link quality. Also the performance of active control algorithm was evaulated.

The result device came in as a complete structure, solving the issues that were to be solved, each part, electical, firmware, mechanics and the optics are cooperating with each other in a intended way. It proves that using custom 3D printed mechanics with off the shelf components it is possible to create operational FSO communication link.

7.2 Future Work

The FSO transceiver and the link are tasks that can be performed in various ways, depending on the needed application. The transceiver described by this document aims to be applied in "Last mile" scenarios.

In such a use case used components, such as SFP modules, are sufficient to connecting two buildings together. Although the low data refresh in DDM technology bottlenecks the possibilities of compensating different environmental factors. With current application the link is capable of reducing the slow paced changes, like thermal expansion of the buildings and the device itself. In higher distances this might start to be an issue where those fast paced vibrations or wind swings might lead to big packet losses. So the first improvement to make such transceiver greater, would be obtaining its own circuit of receiver and transmitter, which would allow for faster readouts of received power.

Another thing that could be upgraded are mechanics; more precisely, the 3d printed guides would be changed with proper precise guide rail, which would eliminate the loose movements and decrease the friction. Current movements of the transceiver are also far too long in tens of centimeters, limited to one centimeter range by the end switches. Limiting it by design will reduce the amount of materials used in production, including filament and lead screw. The production time would also be reduced by shorter 3D print times. The last thing regarding the mechanics would be the sealed waterproof casing of the transceiver, which would allow for proper application.

The last thing that would improve the device would be beacon light source, which would allow for link set up without human interaction. Such light paired with couple photodiodes would make the setup faster but also would open the new possibilities of automatic link recovery in case of total link degradation.

In the end, it is a Proof of Concept device that is shown to fulfill the objectives. Previously mentioned improvements might be used to improve this device in real world applications.

Bibliography

- [1] Mojtaba Mansour Abadi, Zabih Ghassemloooy, Nithin Mohan, Stanislav Zvanovec, Manav R Bhatnagar, and Ralph Hudson. Implementation and evaluation of a gigabit ethernet fso link for'the last metre and last mile access network'. In *2019 IEEE International Conference on Communications Workshops (ICC Workshops)*, pages 1–6. IEEE, 2019.
- [2] Mazin Ali Abd Ali and Miami Abdulatteef Mohammed. Effect of atmospheric attenuation on laser communications for visible and infrared wavelengths. *Al-Nahrain Journal of Science*, 16(3):133–140, 2013.
- [3] Govind P Agrawal. *Fiber-optic communication systems*. John Wiley & Sons, 2012.
- [4] Nitin Arora, Garima Bhasin, and Neha Sharma. Two way linear search algorithm. *International Journal of Computer Applications*, 107(21), 2014.
- [5] M Arumugam. Optical fiber communication—an overview. *Pramana*, 57(5):849–869, 2001.
- [6] AscentOptics. Can you read wdm in a minute. <https://ascentoptics.com/blog/pl/can-you-read-wdm-in-a-minute/>, 6 2023. [Accessed: 2026-06-24].
- [7] AscentOptics. What is dwdm? technology, benefits and applications. <https://ascentoptics.com/blog/pl/what-is-dwdm/>, 3 2026. [Accessed: 2026-06-24].
- [8] Scott Bloom et al. The physics of free-space optics. *AirFiber Inc. White Paper*, pages 1–22, 2002.
- [9] Sathiyaraj Chinnasamy, M Ramachandran, M Amudha, and Kurinjimalar Ramu. A review on hill climbing optimization methodology. *Recent trends in management and commerce*, 3(1):1–7, 2022.
- [10] Artur Czerwinski, Mikołaj Lasota, Marcin Jarzyna, Mateusz Kucharczyk, Michał Jachura, and Konrad Banaszek. Modeling optical key distribution over a satellite-to-ground link under weak atmospheric turbulence. *arXiv*, 2025.
- [11] D. Driscoll, B. Zellers, J. Schomacker, J. Carro, W. T. Roberts, D. MacDonald, J. Guregian, and W. Klipstein. Deep space optical communications (DSOC) beam expander design and engineering. In *Free-Space Laser Communications XXXII*, volume 11272, page 112720H. SPIE, 2020.
- [12] Hennes Henniger and Otakar Wilfert. An introduction to free-space optical communications. *Radioengineering*, 19(2):203–212, 2010.
- [13] Rongqing Hui. *Introduction to fiber-optic communications*. Academic Press, 2025.
- [14] Joshua Ingersoll. A technical evaluation of integrating optical inter-satellite links into proliferated polar leo constellations. In *Proceedings of the 34th Annual AIAA/USU Conference on Small Satellites*. Utah State University, Aug 2020.

- [15] iPerf. iperf - the tcp, udp and sctp network bandwidth measurement tool. <https://iperf.fr/>. Accessed: 2026-06-25.
- [16] Mohammad Ali Khalighi and Murat Uysal. Survey on free space optical communication: A communication theory perspective. *IEEE communications surveys & tutorials*, 16(4):2231–2258, 2014.
- [17] Amith Khandakar, Abir Touati, Farid Touati, Abderrazak Abdaoui, and Ammar Bouallegue. Experimental setup to validate the effects of major environmental parameters on the performance of fso communication link in qatar. *Applied Sciences*, 8(12):2599, 2018.
- [18] Linovision Europe Store. What is inside an sfp module? – understanding tosa, rosa, bosa. <https://eu.linovision.com/blogs/blog/what-is-inside-an-sfp-module-understanding-tosa-rosa-bosa>, July 2024. Accessed: 2026-06-24.
- [19] Volodymyr Lotysh, Larysa Gumeniuk, and Pavlo Humeniuk. Comparison of the effectiveness of time series analysis methods: Sma, wma, ema, ewma, and kalman filter for data analysis. *Informatyka, automatyka, pomiary w gospodarce i ochronie środowiska*, 13(3):71–74, 2023.
- [20] Arun K Majumdar. Fundamentals of free-space optical (fso) communication system. In *Advanced free space optics (FSO) a systems approach*, pages 1–20. Springer, 2014.
- [21] Aditi Malik and Preeti Singh. Free space optics: current applications and future challenges. *International journal of optics*, 2015(1):945483, 2015.
- [22] Luka Mustafa, Eva Černič, and Benn Thomsen. Fso artificial low-cost fog attenuation experiment design. In *2016 10th International Symposium on Communication Systems, Networks and Digital Signal Processing (CSNDSP)*, pages 1–5. IEEE, 2016.
- [23] Roberto Pasic, Ivo Kuzmanov, and Kokan Atanasovski. Esp-now communication protocol with esp32. *Journal of Universal Excellence*, 6(1):53–60, 2021.
- [24] Bruno Rădulescu, Andrei Marius Mihalache, Adelina Hrițuc, Mara Rădulescu, Laurențiu Slătineanu, Adriana Munteanu, Oana Dodun, and Gheorghe Nagîț. Thermal expansion of plastics used for 3d printing. *Polymers*, 14(15):3061, 2022.
- [25] Robofiber, Inc., Alamo, CA. *10Gb/s SFP+ Optical Transceiver Module SFP-1000-LRM (Rev. 1.4)*, April 2013. Datasheet.
- [26] Eddie Ruff and Cullen Bradley. Self-healing free space optical link. *Optical Engineering*, 63(4):041210, 2024.
- [27] Storage Networking Industry Association (SNIA). Sff-8472 specification for management interface for sfp+. Specification Rev. 12.4, SNIA SFF Technology Affiliate Technical Work Group, 2021.
- [28] Abderrahmen Trichili, Mitchell A Cox, Boon S Ooi, and Mohamed-Slim Alouini. Roadmap to free space optics. *Journal of the Optical Society of America B*, 37(11):A184–A201, 2020.

- [29] Luanxia Yang, Bingcheng Zhu, Julian Cheng, and Jonathan F Holzman. Free-space optical communications using on–off keying and source information transformation. *Journal of Lightwave Technology*, 34(11):2601–2609, 2016.

A. Appendices

Appendix A: List of elements per transceiver

Tab. A.1: Cost-oriented Bill of Materials for the prototype FSO node.

Component Description	Details / Quantity	Total Cost (PLN)
Stepper Motor	3× NEMA17 + A4988 drivers (from 4-pack)	120.00
Nucleo Board	1x	70.00
ESP32C6 dev module	1x	20.00
Stiff Clutch	3×	20.00
Bearing	3×	~10.00
Anti-backlash Nut (1 mm pitch)	3×	27.00
Lead Screw (1 mm pitch)	1×	50.00
Screws & Nuts	Assorted hardware	~10.00
Optical Lens	D=52 mm, F=80 mm	30.00
SFP Transceiver 1550 / 1330 nm SC DDM 40km+	1x	50.00
Media Converter	TP-Link MC220L	~90.00
PLA Filament (Structural Mechanics)	700 g	~50.00
Total Prototype Cost		~550.00
Total Link cost (2x node)		~1100.00

Appendix B: CAD raport

